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Modeling Cross-Market Dependence of Global Assets via High-Dimensional Copulas

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Abstract

The thesis aims to investigate the dependence structure of global assets returns using truncated vine copula models. The weekly returns of 30 global assets across five major asset classes are modeled using vine copulas, specifically, truncated R-vine and truncated C-vine. The performance of the vine models is evaluated through Value-at-Risk (VaR) backtesting. The results have shown that the truncated C-vine model provides more accurate risk estimates compared to traditional approaches such as the Mean–Variance portfolio and other vine structures.

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1 Introduction

1.1 Background and Motivation

The benefits of global asset allocation have been recognized for decades. By investing globally among multiple asset classes and regions, investors can diversify the risks and thus achieve more stable returns. However, in practice, portfolio risk is often not accurately estimated, especially during periods of crisis when co-movements across assets tend to be more distinct. One of the most important reasons for this shortcoming lies in the limitations of traditional modeling approaches.

The traditional Mean Variance model proposed by Markowitz(Markowitz, 1952) assumes joint normality of asset returns, but empirical evidence (see, e.g. Tsay, 2010) shows that financial returns exhibit excess kurtosis and heavy tails, leading to underestimated tail risks. Moreover, its reliance on linear correlation measures fails to capture complex dependence structures, particularly during periods of market crisis when co-movements across assets tend to strengthen.

Copulas provide a solution to address the problems stated above and have been extensively studied for decades. Copula models, especially the Vine copula models offer flexibility in modeling by allowing the marginal distributions and dependence structures to be modeled separately. This framework is particularly well-suited for capturing non-linear relationship and tail dependence.

However, even though the Vine copula have been well developed and extensively applied in the past decade, the empirical applications in finance with these Vine copulas rarely exceed 20 dimensions. For instance, in the fundamental paper introducing pair-copula constructions, Aas et al. (2009) illustrated the methodology with examples in three to five dimensions and carried out an empirical study on only five assets. Similarly, Joe (2014) applied the framework to seven-dimensional asset returns.

Even with advances in computation in recent years, most empirical studies remain confined to a single asset class(see, e.g. Czado, 2019). This highlights the need for high-dimensional modeling in global assets among multiple asset groups.

This thesis addresses the empirical gap by modeling the non-linear dependence structure of 30 global assets from five distinct groups, thereby providing a unified framework for analyzing cross-market dependence.

Code and data are available at: https://github.com/lapinou0804/copula-thesis

1.2 Thesis Structure

The thesis is structured as follows: It begins with a presentation of the core underlying statistical and econometric theories which are necessary for the analysis. The subsequent section outlines the methodology in detail. This is followed by empirical analysis, where the proposed methodology is applied to the data. Finally, the thesis concludes with a summary of the main findings and implications.

2 Theoretical Background

2.1 Mean-Variance Portfolio

The classical mean–variance (MV) framework proposed by Markowitz (1952) formulates portfolio selection as the following optimization problem:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^\top \boldsymbol{\mu} \\ \text{s.t.} \quad & \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \leq \sigma_p^2, \\ & \mathbf{w}^\top \mathbf{1} = 1, \\ & \mathbf{w} \geq 0, \end{aligned}$$

where $\mathbf{w} \in \mathbb{R}^n$ denotes the vector of portfolio weights, $\boldsymbol{\mu} \in \mathbb{R}^n$ represents expected returns, and $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$ is the covariance matrix of asset returns, the constraint $\mathbf{w} \geq 0$ implies that no short-selling is allowed.

While this framework provides a foundational approach to portfolio optimization, it relies heavily on the assumption of joint normality and linear dependence among asset returns(Embrechts et al., 2002) . However, empirical evidences of financial data constantly show that asset returns exhibit excess kurtosis and strong non-linear co-movements, particularly during periods of market stress(Tsay, 2010). As a result, the assumption of joint normality and linear dependence may lead to a systematic underestimation of tail risks. In fact, the Markowitz Portfolio have been widely criticized as the main reason for the underestimation of risks which led to the 2008 financial crisis(Salmon, 2009).

2.2 ARMA-GARCH

ARMA–GARCH consists of two time series models: ARMA models the the conditional mean, and GARCH is used to model the conditional variance of a time series. The ARMA part captures linear serial dependence in the return process, while the GARCH part captures time-varying volatility, especially the volatility clustering that is commonly observed in financial data. Using an ARMA-GARCH model to filter the financial data

and transform the standardized residuals into pseudo observations is a standard approach before copula modeling(Tsay, 2010; Francq and Zakoïan, 2019).

Let (r_t) denote a return series. An ARMA(p, q) model can be written as

$$r_t = \mu + \sum_{i=1}^{p} \phi_i (r_{t-i} - \mu) + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t,$$

where (ε_t) is white noise with zero mean and unit variance. The ARMA model explains the current returns in two parts: The AR(auto-regressive) part explains the current return in the form of its past values; while the MA(moving average) part tries to explain the current returns in the form of past shocks. ARMA model is central to classical time-series models and is typically used to describe the conditional mean dynamics (Box et al., 2016; Tsay, 2010).

However, one of the typical features of financial returns is volatility clustering, in other words, large returns are usually followed by large returns and vice versa, this characteristic is particularly distinct during the time of financial stress. Therefore, modeling only the conditional mean is usually not sufficient, as the variance often changes over time. To address this problem, Engle (1982) introduced the ARCH model, in which the conditional variance depends on past squared shocks. A simple ARCH(m) specification is

$$\varepsilon_t = \sigma_t z_t, \qquad z_t \overset{i.i.d.}{\sim} (0, 1),$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{m} \alpha_i \varepsilon_{t-i}^2, (\omega > 0, \alpha_i \geq 0).$$

This model explains that large past shocks increase current conditional variance, which reflects the volatility clustering phenomenon.

Bollerslev (1986) generalized this framework by introducing the GARCH model, in which the conditional variance is explained not only by past squared shocks but also by its own past values. The GARCH(m, s) model is specified as

$$\varepsilon_t = \sigma_t z_t, \qquad z_t \overset{i.i.d.}{\sim} (0, 1),$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{m} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{s} \beta_j \sigma_{t-j}^2.$$

This extension is important because it allows volatility persistence to be modeled more parsimoniously than a high-order ARCH model. In empirical finance, the GARCH$(1, 1)$ specification is often used as a default model (Bollerslev, 1986; Tsay, 2010).

Combining both the auto-regressive and moving-average parts yields the ARMA-GARCH model:

$$r_t = \mu_t + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t,$$

where μ_t follows an ARMA process and σ_t^2 follows a GARCH process. In practice, this means that the mean of returns is modeled in the mean equation, while the volatility around that mean is modeled in the variance equation. In financial applications, the innovation z_t is often assumed to follow a Gaussian, Student-t, or skewed-t distribution, depending on the heaviness and asymmetry of the tails (Tsay, 2010; Francq and Zakoïan, 2019).

2.3 Copula

Sklar's theorem states that any multivariate distribution can be decomposed into its marginal distributions (individual behavior) and a copula that completely captures the dependence structure among the variables (Sklar, 1959). Formally,

Let F be a joint distribution function of a random vector $(X_1, \ldots, X_d)$ with marginal distribution functions $F_1, \ldots, F_d$. Then there exists a copula $C : [0,1]^d \to [0,1]$ such that

$$F(x_1, \ldots, x_d) = C\big(F_1(x_1), \ldots, F_d(x_d)\big).$$

If $F_1, \ldots, F_d$ are continuous, then the copula C is unique. Conversely, for any copula C and marginal distribution functions $F_1, \ldots, F_d$, the function F defined above is a valid joint distribution.

If the joint distribution admits a density, then sklar's theorem can also be written as

$$f(x_1, \ldots, x_d) = c\big(F_1(x_1), \ldots, F_d(x_d)\big) \prod_{i=1}^{d} f_i(x_i),$$

where c is the copula density and f_i are the marginal densities.

Sklar's theorem provides the theoretical foundation for high-dimensional pair-copula constructions.

Elliptical and Archimedean copulas have been extensively studied and became an important part of literature in statistics in the late $90s$. While in two dimensions there is a wide range of copulas available to model different dependence structures, multivariate copula modeling is more challenging. Standard classes such as elliptical and Archimedean copulas exhibit severe drawbacks in higher dimensions. In fact, these copulas cannot correctly model the dependence structure of the real world financial data. For example, Archimedean copulas, which typically have only one or two parameters, impose very strict

dependence properties by assuming exchangeability and that all multivariate margins are the same. Elliptical copulas such as the Gaussian or the Student's t, on the other hand, suffer from a large number of parameters and symmetry restrictions in the tails(Brechmann and Czado, 2013). Additionally, extending the bivariate copulas to the multivariate case is computationally difficult (Nelsen, 2006, p. 105). In order to solve the concerns mentioned above, a new statistical method called pair copula construction and thus vine copula were introduced.

Constructed by pair copulas, vine copulas allow different behaviors of pairs of variables to be modeled suitably, in particular so with regard to their tail asymmetry, or lack thereof, strength of dependence. Such flexibility requires well-designed model selection procedures to realize the full potential of vine copulas as dependence models.

2.3.1 R-Vine

Regular vine (R-vine) provides maximum flexibility in capturing pairwise dependencies among all the vine structures, it is used when the primary goal of the research is purely predictive accuracy, without the need for financial or economic interpretation.

An R-vine copula decompose the multivariate copula density as a product of bivariate copulas, organized in a sequence of trees $T_1, \ldots, T_{d-1}$ that satisfy the proximity condition(Aas et al., 2009). Bedford and Cooke (2002) have introduced the R-vine structure in a graphical model.

Formally, an R-vine on d variables is defined by a sequence of linked trees

$$\mathcal{V} = (T_1, T_2, \ldots, T_{d-1}),$$

where T_1 has node set $N_1 = \{1, \ldots, d\}$, for each $k \geq 2$, the node set of tree T_k is given by the edge set of the preceding tree T_{k-1}. In addition, the sequence of trees must satisfy the proximity condition: two nodes in tree T_k can be connected by an edge only if the corresponding edges in tree T_{k-1} share a common node (Bedford and Cooke, 2002; Aas et al., 2009). The proximity condition is crucial in R-vine construction and guarantees that the resulting multivariate copula is mathematically defined.

The density of an R-vine copula can be written as

$$c(u_1, \ldots, u_d) = \prod_{k=1}^{d-1} \prod_{e \in E_k} c_{j_e, \ell_e | D_e} \left(u_{j_e | D_e}, u_{\ell_e | D_e}; \theta_{j_e, \ell_e | D_e} \right),$$

where, for each edge $e \in E_k$, j_e and ℓ_e denote the conditioned variables, D_e denotes the conditioning set with cardinality $|D_e| = k-1$, and $c_{j_e, \ell_e | D_e}$ is the corresponding conditional

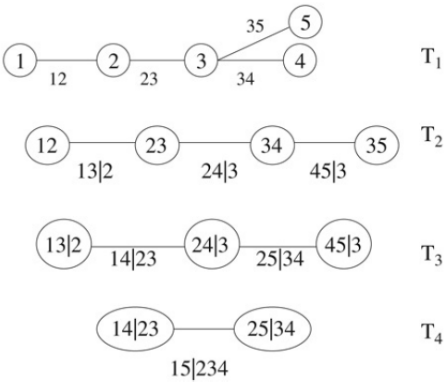

Figure 1: R-vine Tree with Five Variables

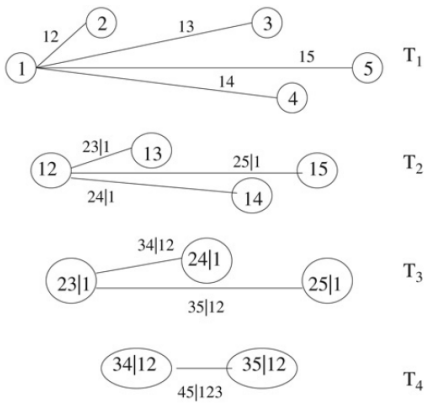

Figure 2: C-vine Tree with Five Variables

pair-copula density with parameter vector $\theta_{j_e,\ell_e|D_e}$ (Aas et al., 2009; Joe, 2014).

A graphic example of R vine consisting of five variables is shown in Figure 1.

2.3.2 C-Vine

Canonical vine (C-vine) is a specific class of regular vines, and fitting a canonical vine might be advantageous when a particular variable is known to be a key variable that governs interactions in the data set. In such a situation one may decide to define this variable as the root of the canonical vine Aas et al. (2009). The root nodes of C-vine trees are determined by identifying the node with strongest dependencies to all other nodes. The most commonly used measure for identifying the strongest node is the absolute value of empirical Kendall's tau whereas other measures include Spearman's rho, AIC or BIC.

A C-vine copula expresses the copula density as a product of conditional bivariate copulas, organized in a sequence of trees $T_1, \ldots, T_{d-1}$, where in each tree one variable (the root) is connected to all others. The density is given by:

$$c(u_1, \ldots, u_d) = \prod_{j=1}^{d-1} \prod_{k=j+1}^{d} c_{j,k|1,\ldots,j-1}\left(u_{j|1,\ldots,j-1},\, u_{k|1,\ldots,j-1}; \theta_{j,k|1,\ldots,j-1}\right),$$

where $c_{j,k|1,\ldots,j-1}$ is the pair copula density between variables j and k, conditional on $\{1, \ldots, j-1\}$; $u_{j|1,\ldots,j-1}$ is the conditional distribution of u_j given $\{1, \ldots, j-1\}$.

Similar to the R-vine, the C-vine can also be represented as a graphical model. Figure 2 illustrates a canonical vine structure with five variables.

2.3.3 Vine Copula Estimation

Joint maximum likelihood estimation of regular vine copula parameters can be very challenging in high dimensions. As a result, we adopt a sequential parametric estimation strategy for our vine copulas, which is the prevailing approach in high-dimensional applications(Joe, 2014; Aas et al., 2009). Let $U = (U_1, \ldots, U_d)$ be the PIT–transformed uniforms. The vine copula density factorizes as

$$c(u_1, \ldots, u_d) = \prod_{k=1}^{d-1} \prod_{(i,j|D)\in\mathcal{E}_k} c_{i,j\,|\,D}\big(u_{i\,|\,D},\, u_{j\,|\,D};\, \theta_{i,j\,|\,D}\big),$$

where $\mathcal{E}_k$ is the edge set of tree k and $u_{i\,|\,D}$ are conditional uniforms obtained via h–functions(see Aas et al., 2009). The sequential estimator proceeds tree by tree. It first estimates all edges in Tree 1 using (U_i, U_j); then compute conditional uniforms $U_{i\,|\,D} = h_{i|D}(\cdot)$ from the fitted edges; then estimate Tree 2 on $(U_{i\,|\,\{v_1\}}, U_{j\,|\,\{v_1\}})$; and so on. This yields consistent estimators under standard regularity conditions and is computationally tractable (Aas et al., 2009; Joe, 2014).

Elliptical copulas and Archimedean copulas are usually used in empirical analysis for model flexibility. The criteria for copula selection will be AIC because in a large-scale Monte Carlo study the AIC turned out to be the most reliable selection criterion (Manner, 2007).

2.4 Value at Risk (VaR)

Value at Risk (VaR) is a popular downside risk measure. The Basel Committee on Banking Supervision(Basel Committee on Banking Supervision, 2019) defines Value at Risk (VaR) as

> the maximum potential loss in the value of a portfolio over a given time horizon at a specified confidence level.

Formally, let L_t denote the portfolio loss at time t. The VaR at confidence level $\alpha \in (0, 1)$ is defined as the α-quantile of the loss distribution:

$$\mathrm{VaR}_\alpha(L_t) = \inf\{x \in \mathbb{R} : \mathbb{P}(L_t \leq x) \geq \alpha\}$$

In practice, there are mainly three statistical approaches to estimate the Value at Risk, namely parametric approach, historical simulation and Monte Carlo simulaitons(J.P. Morgan/Reuters, 1996). The parametric approach assumes a specific distribution for asset returns, typically normal or Student-t, and computes VaR analytically based on estimated moments. Historical simulation estimates VaR non-parametrically using the empirical dis-

tribution of past returns. This thesis will apply Monte Carlo simulation to estimate VaR. The simulation process generates synthetic portfolio returns from the fitted copula model, which allow for greater flexibility in capturing non-linear and tail dependence structures. −VaR is therefore calculated as the empirical quantile of the simulated portfolio's return distribution.

To evaluate the accuracy of VaR forecast, Some test statistics such as Kupiec test for unconditional coverage can be applied.

3 Methodology

Methodologically, the analysis begins with time series modeling to filter the raw weekly return data,since copula models request i.i.d inputs. The dependence structure is then modeled using a multivariate t -copula, as well as R-vine and C-vine copula constructions. To assess model performance, Value-at-Risk (VaR) is employed as the risk metric, evaluated through a rolling-window backtesting procedure.

3.1 Marginal Model Specification

We aim to model the dependency structures of global assets across five different asset groups, namely, equities, fixed income, currencies, commodities, and alternatives. The financial data one can obtain from financial data providers are typically expressed in the form of asset prices. However,in time series analysis, asset prices are generally non-stationary, whereas the returns or log-returns are (Ruppert and Matteson, 2015, pp. 307–309). Thus, it is necessary to transform asset prices P_t into returns before further modeling. The simple return at time t for an asset is defined as

$$R_t = \frac{P_t}{P_{t-1}} - 1$$

Since copulas are inherently static models, they have to be combined with time series models to model financial returns (see, e.g. Cherubini et al., 2004). Copula modeling then proceeds with standardized residuals obtained from these models to describe the dependence structure among assets, allowing the joint distribution to be decomposed into well-specified marginals and a flexible dependence function.

3.1.1 AR(1)-GARCH(1,1)

A widely adopted approach in financial econometrics is to model each return series using an AR(1)–GARCH(1,1) specification with Student-t or skewed Student-t innovations, allowing for both serial dependence in the mean and time-varying conditional heteroskedasticity

with heavy tails. The model is specified as follows:

$$
\begin{aligned}
r_t &= \mu + \phi_1 r_{t-1} + \varepsilon_t, \\
\varepsilon_t &= \sigma_t z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} (0, 1), \\
\sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2
\end{aligned}
$$

Under the correct model specification, the resulting standardized residuals $\hat{z}_{i,t} = \hat{\varepsilon}_{i,t}/\hat{\sigma}_{i,t}$ are expected to be approximately independent and identically distributed (i.i.d.) with zero mean and unit variance.

Some diagnostic checks on the standardized residuals can be performed to assess if the i.i.d assumption is met. The Ljung–Box test is applied to both $\hat{z}_{i,t}$ and their squared values $\hat{z}_{i,t}^2$ to detect any remaining serial correlation in conditional mean and conditional variance, respectively. The absence of significant autocorrelation provides empirical support for the assumption of i.i.d. innovations, which is a crucial prerequisite for subsequent copula-based dependence modeling.

3.1.2 Probability Integral Transform (PIT)

Copula models require inputs to be uniformly distributed on the $[0, 1]$ interval, yet the standardized residuals obtained from marginal time series models do not generally satisfy this requirement. Therefore, to prepare inputs suitable for copula modeling, the probability integral transform (PIT) is finally applied to the standardized residuals

$$u_{i,t} = \hat{F}_i(\hat{z}_{i,t}),$$

where $\hat{F}_i$ is the estimated cumulative distribution function of the innovation distribution for asset i. By construction, the transformed series $(u_{i,t})$ are approximately uniform on $[0, 1]$ and can therefore be served as inputs for copula estimation.

3.2 Dependence Modeling Using Copulas

3.2.1 Multivariate t- copula

The n-dimensional Student copula has been used repeatedly for modeling multivariate financial return data. A number of papers, such as Mashal and Zeevi (2002), have shown that the fit of this copula is generally superior to that of other n-dimensional copula for such data. Thus, it is a good choice to adopt t- copula as our baseline model for the 30 dimensional global asset returns.

However, the Student copula has only one parameter for modeling tail dependence, independent of dimension. Hence, if the tail dependences of different pairs of variables in a

portfolio are very different, which is the case in cross market assets, a better description of the dependence structure can be achieved with other methods such as the pair copula decomposition with C, D, or R vines. More details can be found in (Aas et al., 2009).

3.2.2 Truncated R-Vine

As mentioned in subsubsection 2.3.1, an R-vine on d variables is defined by a sequence of linked trees. Therefore, the first step in fitting R-vine copula is to determine the trees

$$\mathcal{V} = (T_1, T_2, \ldots, T_{d-1}).$$

Tree 1 is constructed by means of a maximum spanning tree (MST) based on empirical Kendall's tau. For each pair of variables (i, j), the edge weight is defined as

$$w_{ij} = \left|\hat{\tau}_{ij}\right|,$$

where $\hat{\tau}_{ij}$ denotes the empirical Kendall's tau between the PIT transformed pseudo-observations U_i and U_j. Let $\mathcal{T}_d$ denote the set of all spanning trees on the node set $\{1, \ldots, d\}$. Then Tree 1 is selected as

$$T_1 = \arg\max_{T \in \mathcal{T}_d} \sum_{(i,j) \in E(T)} w_{ij},$$

where $E(T)$ denotes the edge set of tree T. This criterion ensures that the strongest unconditional pairwise dependencies are captured in the first tree. Higher-order trees $T_2, \ldots, T_{d-1}$ are then constructed recursively with the same method as Tree 1.

The sequential estimation will be applied to truncated R-vine and the copula family will be restricted to elliptical copulas and Archimedean copulas for model flexibility.

Truncation However, for d variables, there are $d(d-1)/2$ pair copulas and the number of parameters increases drastically as the dimension increases. For example, an R vine structure with n variables with bivariate $t$copula (two-parametric copula) has $n(n-1)$ parameters(Czado et al., 2013). This may result in a huge computational burden. Hence, it is necessary to reduce the number of parameters by truncating the vine structure.

For R-vine models, a bottom-up selection approach is employed. Trees are added sequentially until the model fit, measured by the Bayesian Information Criterion (BIC) no longer improves (Nagler and Czado, 2016). Specifically, the procedure stops when $\Delta\text{BIC} > 0$.

The algorithm for selecting the truncation level is as follows:

1. Initialize $k = 1$.

2. Fit the k-th tree T_k of the R-vine model.

3. Compute the Bayesian Information Criterion $\mathrm{BIC}(k)$ for the model including trees up to level k.

4. Evaluate the change in BIC:

$$\Delta\mathrm{BIC} = \mathrm{BIC}(k) - \mathrm{BIC}(k-1).$$

5. If $\Delta\mathrm{BIC} > 0$, stop the procedure. The optimal truncation level is $k-1$, and all higher-order trees $T_k, \ldots, T_{d-1}$ are replaced by independence copulas.

6. Otherwise, increment $k \leftarrow k+1$ and return to Step 2.

3.2.3 Truncated C-Vine

Since the assets from five asset groups display a natural hierarchical structure, C vine copula appears to be an advantageous model to obtain an economically interpretable dependence, thus we specify a truncated C-vine copula with a pre-defined variable ordering.

To construct a hierarchical Canonical vine, the variables should be ordered in a way such that the first nodes in the vine correspond to the most important variables in the dependence structure.

There are several possible ways to determine the variable ordering of a C-vine. In this thesis, the ordering is based on empirical Kendall's tau. The first root node, referred to as the global root or global hub, is selected as

$$v_1 = \arg\max_{i\in\{1,\ldots,30\}} \sum_{\substack{j=1\\ j\neq i}}^{30} \left|\hat{\tau}_{ij}\right|,$$

where $\hat{\tau}_{ij}$ denotes the empirical Kendall's tau between assets i and j. Hence, v_1 is interpreted as the asset that is most strongly connected to the rest of the system, and it therefore serves as the root node in Tree 1 of the C-vine structure.

After the global root v_1 is selected, one representative is chosen from each asset group $\mathcal{G}_g$. Specifically, for each class g, the class representative is defined by

$$r_g = \arg\max_{i\in\mathcal{G}_g,\, i\neq v_1} \sum_{\substack{j\in\mathcal{G}_g\\ j\neq i}} \left|\hat{\tau}_{ij}\right|.$$

These five class representatives are then ranked according to their overall global depen-

dence score

$$S(r_g) = \sum_{\substack{j=1 \\ j \neq r_g}}^{30} \left| \hat{\tau}_{r_g j} \right|,$$

and the ordered representatives are assigned to positions $v_2, \ldots, v_6$ in decreasing order of $S(r_g)$.Then sort these five representatives from largest to smallest $S(r)$.

The final ordering of the C-vine is thus

$$(v_1, v_2, v_3, v_4, v_5, v_6, \ldots, v_{30}),$$

where v_1 is the global root, $v_2, \ldots, v_6$ are the five ordered asset-class representatives, and $v_7, \ldots, v_{30}$ are the remaining assets, appended class by class and ordered by decreasing within-class dependence strength.

Truncation: For C-vines, a natural truncation strategy in practice is to fix the maximum tree depth K, i.e. to estimate only the first K trees and set higher-order conditional copulas to independence. This approach has been recommended in high-dimensional applications (Czado et al., 2013) and is implemented as standard in the VineCopula R package. Given that the empirical objective of the thesis is to capture the dominant dependence structure across the asset groups rather than those high-order conditional interactions, the C-vine is truncated at level $K = 6$, meaning all pair-copulas beyond Tree 6 are set to independence. This yields a parsimonious and economically interpretable dependence model for the high-dimensional portfolio.

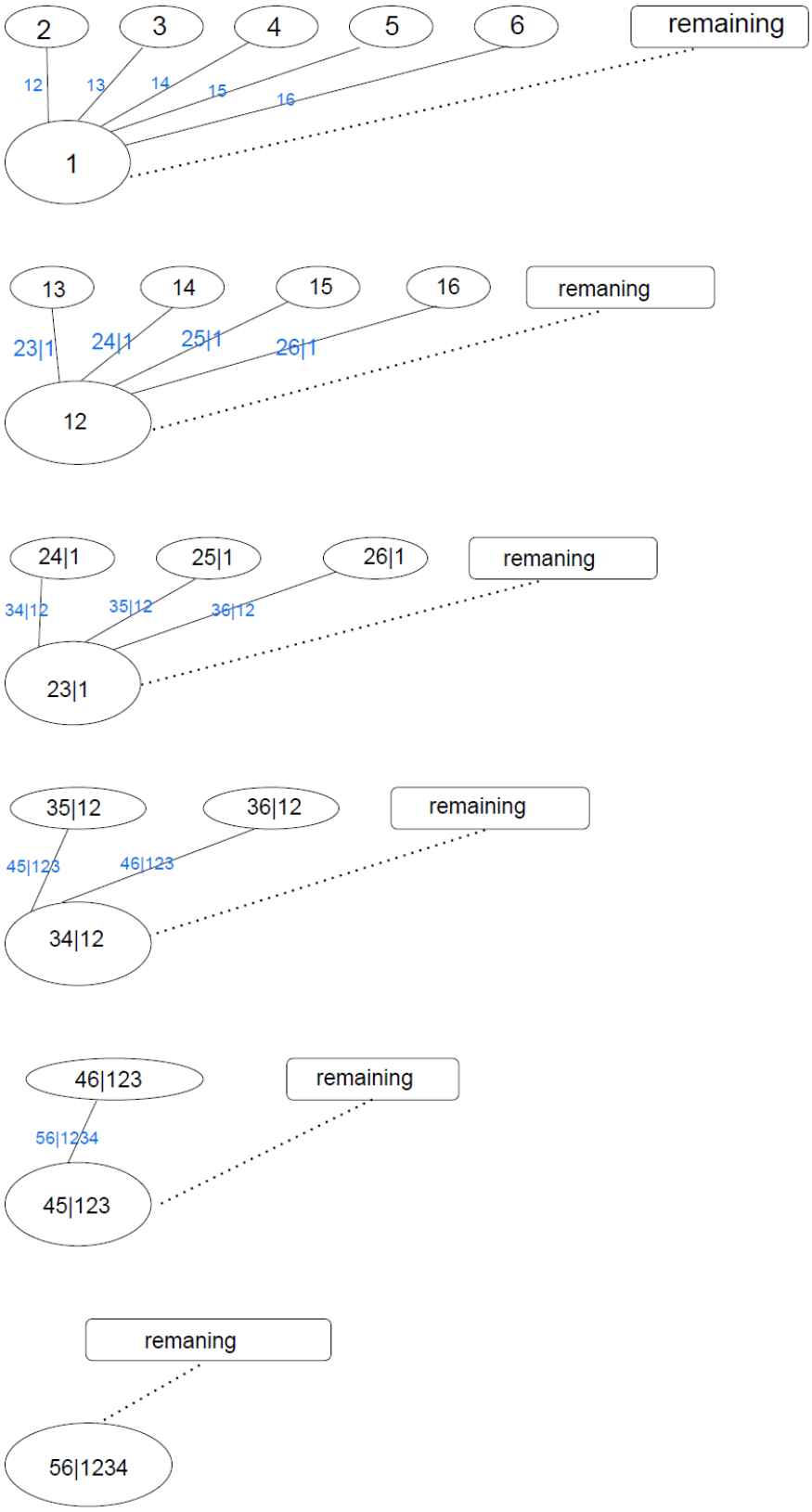

Figure 3: Truncated C-vine design

A schematic design of the truncated C-Vine structure for the 30 assets is shown in Figure 3. From the graphical illustration, we note that Tree 1 has root node v_1 and contains the unconditional pair-copulas $c_{v_1,v_j}, j = 2, \ldots, 30$. Hence, Tree 1 captures the direct dependence between the global root and the other 29 assets.

Tree 2 has root node v_2 and contains conditional pair-copulas of the form $c_{v_2,v_j|v_1}, j = 3, \ldots, 30$. Therefore, this tree captures the residual dependence between v_2 and the remaining assets after conditioning on the first root node v_1 .

Analogously, Trees 3, 4, 5 have root nodes v_4, v_5, and v_6, respectively, with pair-copulas

$$c_{v_k,v_j|v_1,\ldots,v_{k-1}}, \qquad j = k+1, \ldots, 30, \quad k = 3, 4, 5, 6.$$

These trees are designed to capture the dominant cross-class dependence of the global assets in a hierarchical manner.

The sequential estimation will be applied to truncated C-vine as well and the copula family will be restricted to elliptical copulas and Archimedean copulas.

3.3 Simulation

Simulation can be performed after the dependence structure has been modeled using vine copulas. A general simulation algorithm is proposed by Aas et al. (2009), following the pair-copula construction theory. This simulation algorithm allows one to sample d-dimensional dependent variables uniformly distributed on $[0, 1]$, from a given C vine or R vine structure.

In this thesis, the goal is to generate $d = 30$ dependent variables from the fitted R-vine or C-vine models. The simulation procedure can be summarized as follows.

First, generate independent random variables $w_1, \ldots, w_d \sim$ i.i.d. $U(0, 1)$. Then construct the dependent variables recursively:

$$\begin{aligned}
x_1 &= w_1, \\
x_2 &= F^{-1}(w_2 \mid x_1), \\
x_3 &= F^{-1}(w_3 \mid x_1, x_2), \\
&\vdots \\
x_d &= F^{-1}(w_d \mid x_1, \ldots, x_{d-1}).
\end{aligned}$$

The conditional distribution functions $F(x_j \mid x_1, \ldots, x_{j-1})$ are computed recursively using the so-called h-functions, as introduced in Aas et al. (2009). These functions allow the

evaluation of marginal conditional distributions within R-vine or C-vine structures..

Likewise, the simulation algorithm in the copula space can be explained as follows

Let $w_1, \ldots, w_d \sim$ i.i.d. $U(0,1)$. The dependent uniform variables $u_1, \ldots, u_d$ are generated recursively as follows:

$$
\begin{aligned}
u_1 &= w_1, \\
u_2 &= C_{2|1}^{-1}(w_2 \mid u_1), \\
u_3 &= C_{3|1,2}^{-1}(w_3 \mid u_1, u_2), \\
&\vdots \\
u_d &= C_{d|1,\ldots,d-1}^{-1}(w_d \mid u_1, \ldots, u_{d-1})
\end{aligned}
$$

3.4 VaR Backtesting and Performance Evaluation

To assess the performance of the copula model, rolling window backtesting is used. Value at Risk is chosen to be the performance metrics. At each time step, the model is re-estimated using a fixed-size historical window, and a one-step-ahead VaR forecast is thus modeled. The forecast is then compared to the realized portfolio return to determine whether a violation has occurred, i.e., whether the realized loss exceeds the predicted VaR. The sequence of violations is used to assess the accuracy of the model, with particular emphasis to whether the observed number of violations align with the nominal confidence level.

For each rolling window, after fitting the copula and get its corresponding estimated parameters, we are able to simulate one-step-ahead dependent uniforms from the simulation algorithm introduced in the previous section, denoted as $(U_{1,t+1}^{(b)}, \ldots, U_{d,t+1}^{(b)}) \sim \hat{C}, b = 1, \ldots B$, where $\hat{C}$ denotes the fitted vine copula and b is the number of simulations we conduct.

However, these simulated uniforms are pseudo-inputs, not directly the returns. Thus we need to then transform back into simulated standardized innovations via the inverse CDF of the marginal innovation distribution $z_{i,t+1}^{(b)} = F_{Z_i}^{-1}\left(U_{i,t+1}^{(b)}\right)$, where $i = 1, \ldots, d$ is the asset dimension, $t+1$ is the one-step-ahead time point, and $b = 1, \ldots, B$ denotes the simulation draws.

Given the one-step-ahead conditional mean forecast $\hat{\mu}_{i,t+1}$ and conditional volatility forecast $\hat{\sigma}_{i,t+1}$ from the marginal AR(1)-GARCH(1,1)-t model, the simulated asset return is obtained as $R_{i,t+1}^{(b)} = \hat{\mu}_{i,t+1} + \hat{\sigma}_{i,t+1} z_{i,t+1}^{(b)}$

Finally, in order to get the simulated portfolio return, denote $w_1, \ldots, w_d$ the weights fir each asset in the portfolio. For each simulation b, the corresponding simulated portfolio

return is $R_{p,t+1}^{(b)} = \sum_{i=1}^{d} w_i R_{i,t+1}^{(b)}$

However, by obtaining only the simulated portfolio return is not enough to evaluate the performance of the vine structure. Since our main filed of interest is managing financial risks, Value at Risk is chosen to be the risk metrics to evaluate model performance. The reason VaR is suitable for this study is not only because it is a common risk metrics, but its quantile-based nature also makes it well suited for evaluating the tail behavior implied by different dependence models.

Once the simulated portfolio returns $R_{p,t+1}^{(b)}, b = 1, ..., B$ are obtained, the one-step-ahead 95% or 99% fore-casted Value-at-Risk is the empirical 0.05 or 0.01 quantile of this portfolio return distribution. Since portfolio losses correspond to the left tail of the return distribution, the predicted VaR is typically negative. However, the Basel Committee requires it to be reported as a positive value.

A VaR violation occurs whenever the realized portfolio return falls below the predicted VaR threshold. Define the violation indicator

$$I_{t+1} = \mathbf{1}\left\{R_{p,t+1}^{\text{real}} < -\widehat{\text{VaR}}_{0.95,t+1}\right\}, \quad I_{t+1} = \mathbf{1} \text{ if violation}, \quad I_{t+1} = \mathbf{0} \text{ otherwise}.$$

If the copula model accurately predicts the VaR at 95% level, then the empirical average of these indicators should be close to 0.05. Mathematically speaking,

$$\mathbb{E}[I_t] = 0.05 \quad \text{and hence} \quad \frac{1}{T - t_0} \sum_{t=t_0+1}^{T} I_t \approx 0.05.$$

If the empirical average of the violations is much more than 5%, the model underestimates the risks. On the contrary, if the empirical average is much less than 5% then the model is assumed to be too conservative, which overestimates the tail risks.

To test whether the empirical VaR violation is consistent with the nominal level, the Kupiec unconditional coverage test is used(Kupiec, 1995). Let x denote the number of VaR violations over N out-of-sample forecasts, and let α denote the nominal violation probability. The null hypothesis $H_0 : p = \alpha$ is tested against the alternative $H_1 : p \neq \alpha$, where p is the true violation probability. The likelihood ratio statistic follows asymptotically a χ^2 distribution with one degree of freedom:

$$LR_{\text{uc}} = -2\ln\left(\frac{(1-\alpha)^{T-x}\alpha^x}{(1-\hat{p})^{T-x}\hat{p}^x}\right), \qquad \hat{p} = \frac{x}{T}.$$

Failure to reject the null means that the empirical violation frequency is consistent with

the nominal VaR level.

4 Empirical Analysis

All the code and data that are used for empirical analysis are available at: https://github.com/lapinou0804/copula-thesis

4.1 Data Representation and Preliminary Analysis

To ensure consistency with real-world trading conditions, the portfolio of assets for empirical analysis is constructed using highly representative and liquid exchange-traded funds (ETFs) from both developed and emerging markets. Five asset classes are included: equities, fixed income, commodities, currencies, and alternative assets. The price data are obtained from the reliable data provider Yahoo Finance.

The sample period is constrained by data availability, particularly in the fixed income category, where many widely traded ETFs were introduced after 2007. Our dataset covers the period from March 7, 2008 to the last Friday of 2025, with a total number of 928 observations for each asset.

Weekly data is used in this study. While higher-frequency data (e.g., daily returns) may contain richer information, they are often affected by short-term noise, thus leading to higher variance. In contrast, monthly data presents less noise but tend to smooth extreme event, leading to higher bias in estimations. As a result, weekly data is chosen as a balance in between. Moreover, the total number of 928 weekly data observations can also ensure the stability of time series modeling and copula estimation, particularly in a high-dimensional setting.

We denote our five asset groups

$$\mathcal{G}_1 = \{\text{equities}\}, \quad \mathcal{G}_2 = \{\text{fixed income}\}, \quad \mathcal{G}_3 = \{\text{commodities}\},$$

$$\mathcal{G}_4 = \{\text{currencies}\}, \quad \mathcal{G}_5 = \{\text{alternatives}\}$$

More specifically, the column specification the asset matrix is denoted by

$$\mathcal{G}_1 = \{1-10\} \quad \mathcal{G}_2 = \{11-16\} \quad \mathcal{G}_3 = \{17-21\} \quad \mathcal{G}_4 = \{22-26\} \quad \mathcal{G}_5 = \{27-30\}$$

A comprehensive list of the selected assets is shown in Table 1.

Table 1: 30-Asset Portfolio (Mar 2008–Dec 2024)

Asset Class	Yahoo Ticker	Asset Name	Description
Equities	SPY	SPDR S&P 500 ETF Trust	US large-cap benchmark
	QQQ	Invesco QQQ Trust	US tech/growth stocks
	IWM	iShares Russell 2000 ETF	US small-caps
	VGK	Vanguard FTSE Europe ETF	European equities
	EWJ	iShares MSCI Japan ETF	Japanese equities
	EEM	iShares MSCI Emerging Markets ETF	Broad EM exposure
	FXI	iShares China Large-Cap ETF	China equities (launched 2004)
	EWZ	iShares MSCI Brazil ETF	Brazilian equities
	EPI	WisdomTree India Earnings Fund	Emerging market equity (India)
	EFA	iShares MSCI EAFE ETF	Developed markets ex-US & Canada
Fixed Income	IEF	iShares 7-10 Year Treasury Bond ETF	US medium-term Treasuries (data since 2002)
	TLT	iShares 20+ Year Treasury Bond ETF	US long-term Treasuries (data since 2002)
	LQD	iShares iBoxx $ Inv Grade Corp Bond ETF	US corporate bonds (data since 2002)
	JNK	SPDR Bloomberg High Yield Bond ETF	US high-yield bonds (launched 2007)
	EMB	iShares J.P. Morgan USD EM Bond ETF	EM sovereign debt (launched 2007)
	PCY	Invesco Emerging Markets Sovereign Debt ETF	Alternative EM bonds (launched 2007)
Commodities	GC=F	Gold Futures	Precious metal benchmark
	CL=F	Crude Oil WTI Futures	Energy benchmark
	NG=F	Natural Gas Futures	Energy commodity
	SI=F	Silver Futures	Precious/industrial metal
	DBA	Invesco DB Agriculture Fund	Broad agriculture basket
Currencies	EURUSD=X	Euro/US Dollar	Major currency pair
	JPYUSD=X	Japanese Yen/US Dollar	Asian currency
	GBPUSD=X	British Pound/US Dollar	European currency
	AUDUSD=X	Australian Dollar/US Dollar	Commodity currency
	CNYUSD=X	Chinese Yuan/US Dollar	EM currency
Alternatives	VNQ	Vanguard Real Estate ETF	US REITs
	GLD	SPDR Gold Shares	Physical gold
	VXF	Vanguard Extended Market ETF	US mid/small caps
	IYR	iShares U.S. Real Estate ETF	Alternative real estate

To calculate the weekly returns from the raw price data, the followed formula is applied

$$R_t^{(w)} = \left(\frac{\text{Adj Close}_t - \text{Adj Close}_{t-1}}{\text{Adj Close}_{t-1}} \right)$$

where Adj Close$_t$ denotes the adjusted closing price at week t. This transformation yields a $927 * 30$ matrix of weekly returns, which serves as the input for subsequent time series modeling.

After fitting AR(1)-GARCH(1,1) model, we conduct a Ljung-Box test passes for residuals and squared residuals with lag= 20. All assets passed the test at the 5% significance level, with the exception of silver futures. However, after generating the ACF plot of this asset, it's considered more likely to be a noise rather than systematic ARCH effects.

After fitting the AR(1)-GARCH(1,1) model, Ljung–Box tests are conducted on both residuals and squared residuals with a lag of 20. The rule of thumb to determine the lag can be found in(e.g. Harvey, 1990). At the 5% significance level, the null hypothesis of no serial correlation is not rejected for all assets except silver futures. However, after closely checking the ACF of the residuals, it can be concluded that it is more like to be random fluctuations rather than persistent ARCH effects.

It it also important to check of the standardized residuals follow the Student- t assumption. We compare the theoretical quantiles of t distribution to the empirical quantiles, the graph of four representative assets (IEF, TLT, DBA, and EURUSD.X) are presented in Figure 4. Although the empirical tails deviate slightly from the theoretical quantiles, the general fit remains satisfactory. Moreover, Kim et al. (2007) discovered that it would only be a problem when the marginal distributions is severely misspecified.

The probability integral transform (PIT) is therefore applied using a parametric t-distribution, completing the marginal modeling stage and producing approximately i.i.d. uniform variables for copula estimation.

4.2 Multivariate t-copula

The estimated degree of freedom of the multivariate t is $\hat{\nu} = 15.56$, suggesting moderately large tail behaviors, implying that joint extreme events occur more frequently than under a Gaussian copula but are still not extremely significant.

The correlation matrix suggest many values between $0.7 - 0.98$, indicating strong positive co-movement between assets, particularly between the assets within the same group. Some moderate negative correlations also exist, reflects hedging relationships across some asset groups.

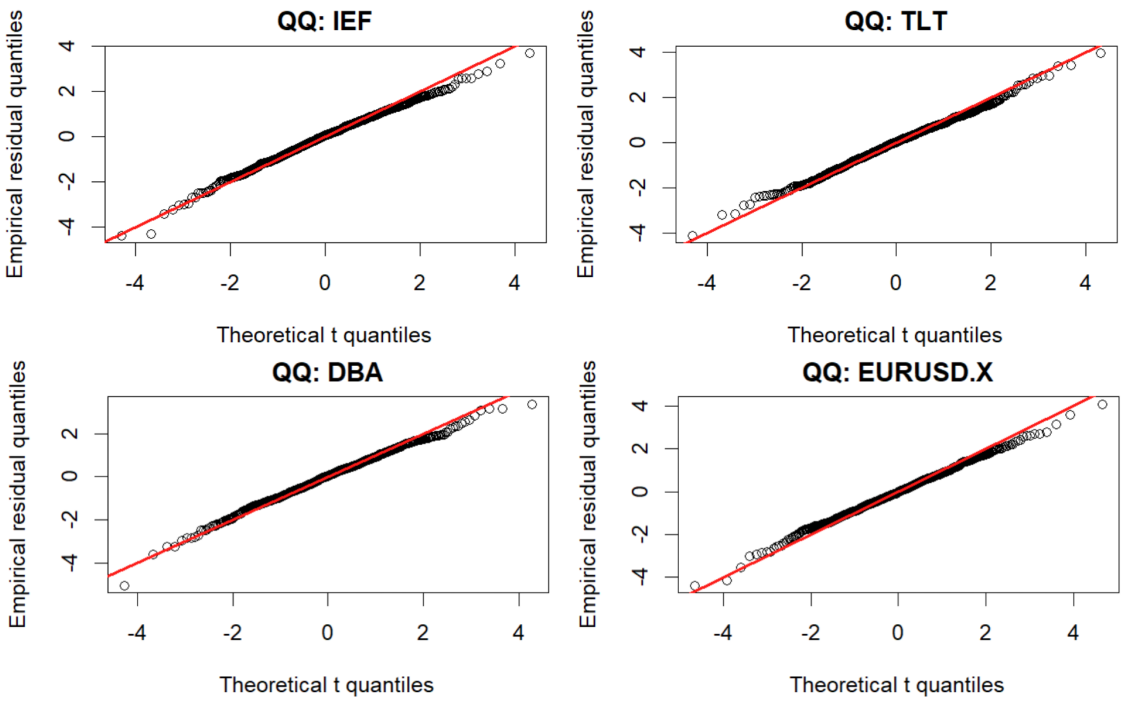

Figure 4: QQ-plots of the standardized residuals against the theoretical Student-t quantiles

However, as stated in the previous section, the multivariate t-copula assumes symmetric dependence and same tail behavior for all pairs of assets, which is unrealistic in the real financial world. What is more, in our analysis, the number of parameters need to be estimated is $435 + 1(df) = 436$, which is also very computationally intensive. These limitations motivate the use of vine copula models, which allow for flexible pairwise dependence modeling.

4.3 R-vine

Table 2: Incremental BIC changes for the first seven vine trees

k	2	3	4	5	6	7	8
ΔBIC	-1437.56	-307.71	-335.60	-106.05	-46.56	-6.16	29.68

Table shows that the incremental change in BIC becomes positive at tree level $k = 8$. According to the algorithm designed in subsubsection 3.2.2, the optimal truncation level is therefore $K = 7$, since adding the $8th$ tree no longer improves the model fit.

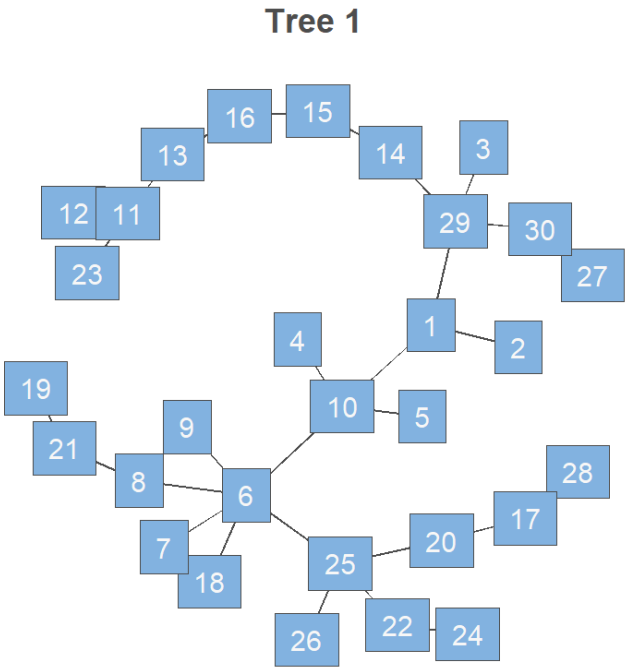

Figure 5: R-vine Tree 1 Structure

The above is a graphical representation of the Tree 1 structure drawn from R -vine. Due to the limit of space, the structure is represented in the form of asset index instead of asset names. The asset names correspond to the list above.

The first tree of the fitted R-vine indicates a clustered dependence structure, though it is not really like a C-vine structure because there are several roots rather than a single dominant root in this tree. There are several node behaving like hubs, especially 6 (EEM), 1(SPY), 25(AUD), 29(VXF), and 10(EFA). These hubs mainly connect the assets in their own asset groups. This suggests that the overall dependence pattern is not driven by a single asset, but rather by several influential assets that organize the dependence structure in different parts of the portfolio.

The flexibility in R-vine is useful in our data analysis because the dependence structure is decentralized, different subsets of assets are driven by different dominant variables. Therefore, the Tree 1 graph provides visual evidence that the dependence structure in the data is complex and clustered rather than purely hierarchical like C-vine or purely chain like D-vine around one single asset.

Table 3 and Table 4 presents the results for the first two trees of the truncated R-vine, including, for each edge, the copula family selected based on the AIC criterion, the estimated parameters for each copula, and the corresponding empirical Kendall's tau values. The results show that Trees 1 and 2 are composed predominantly of t-copulas. In Tree 1, all Kendall's tau values are positive, most of the values are larger than 0.5, thus

Table 3: R-vine results for Tree 1

Pair	Family	par	par2	τ
(6,18)	t	0.43	5.73	0.28
(21,19)	Frank	1.17	–	0.13
(8,21)	t	0.42	13.86	0.28
(6,8)	t	0.79	3.32	0.58
(10,5)	t	0.87	7.16	0.67
(6,7)	t	0.89	9.20	0.70
(10,4)	t	0.98	12.12	0.89
(6,9)	t	0.76	7.66	0.55
(22,24)	t	0.72	11.33	0.51
(25,26)	Frank	3.21	–	0.32
(25,22)	t	0.64	7.24	0.44
(17,28)	t	0.99	3.82	0.91
(20,17)	t	0.85	13.64	0.65
(25,20)	t	0.50	6.42	0.34
(29,3)	t	0.98	4.98	0.88
(6,25)	t	0.63	5.59	0.43
(30,27)	t	1.00	4.33	0.94
(1,2)	t	0.93	8.92	0.76
(10,6)	t	0.86	13.69	0.65
(1,10)	t	0.87	4.78	0.68
(11,23)	t	0.53	4.18	0.36
(11,12)	t	0.93	5.45	0.76
(13,11)	t	0.81	3.08	0.60
(16,13)	t	0.71	3.04	0.51
(15,16)	t	0.97	2.43	0.85
(14,15)	t	0.79	5.19	0.58
(29,1)	t	0.92	5.07	0.75
(29,14)	t	0.79	7.79	0.58
(30,29)	t	0.74	3.97	0.53

Table 4: R-vineresults for Tree 2

Pair	Family	par	par2	τ
(8,18;6)	Gaussian	0.15	–	0.10
(8,19;21)	Survival Gumbel	1.03	–	0.03
(6,21;8)	Frank	0.50	–	0.05
(7,8;6)	Gaussian	-0.27	–	-0.18
(4,5;10)	t	-0.77	9.66	-0.56
(9,7;6)	Gaussian	-0.22	–	-0.14
(6,4;10)	Rotated Gumbel 90°	-1.12	–	-0.11
(10,9;6)	Frank	1.01	–	0.11
(25,24;22)	t	0.32	10.83	0.21
(22,26;25)	Frank	1.31	–	0.14
(20,22;25)	t	0.16	8.67	0.10
(20,28;17)	t	0.00	12.20	0.00
(25,17;20)	t	0.00	7.91	0.00
(6,20;25)	t	0.20	7.10	0.13
(1,3;29)	t	-0.24	5.15	-0.16
(10,25;6)	t	0.19	6.28	0.12
(29,27;30)	t	-0.04	5.29	-0.02
(10,2;1)	t	-0.11	9.69	-0.07
(1,6;10)	t	0.12	9.43	0.08
(29,10;1)	t	0.19	8.98	0.12
(12,23;11)	Rotated Gumbel 270°	-1.04	–	-0.04
(13,12;11)	t	0.22	9.15	0.14
(16,11;13)	t	-0.23	5.14	-0.15
(15,13;16)	t	0.10	5.58	0.06
(14,16;15)	t	0.02	10.12	0.01
(29,15;14)	t	-0.03	13.10	-0.02
(14,1;29)	t	0.21	10.78	0.14
(30,14;29)	t	0.25	6.95	0.16

suggesting relatively strong dependence among the connected asset pairs. One possible reason that all of the tau's values are positive is that the R-vine structure selection procedure prioritizes the strongest pairwise dependencies in the first tree, and in this empirical study, these strongest links are all positive. From Tree 2 onward, negative relationships begin to appear, while the value of Kendall's tau decreases substantially, suggesting that the dependence structure becomes much weaker.

The detailed results for Trees 3 to 6 are reported in the Appendix.

4.4 C-vine

By applying the methodology designed in subsubsection 3.2.3 for truncated C-vine to the empirical analysis, EFA from the equity group is found to have the biggest value of absolute Kendall's tau among all, thus serves as the global hub v_1. Following the same methodology, roots v_2 to v_6 are given in Table 5.

Table 5: Ordered class representatives from v_2 to v_6

Name	Class	Index	WithinClassScore	GlobalScore
EEM	Equity	6	4.2490488	9.634346
IYR	Alternatives	30	1.5188944	7.556182
SI.F	Commodities	20	0.9645784	5.442972
EURUSD.X	Currencies	22	1.3114788	5.058378
LQD	FixedIncome	13	2.1839115	4.602471

Table 6 and Table 7 reports the results for the first two trees of the truncated C-vine, including the copula family selected for each edge based on the AIC criterion, the corresponding parameter estimates, and the empirical Kendall's tau values. The results for Tree 1 are similar to those obtained for the truncated R-vine: The edges are dominated by t-copulas, and the Kendall's tau values are mostly positive, indicating relatively strong dependence among the connected asset pairs. However, Tree 2 of the truncated C-vine shows greater diversity in the selected copula families than the corresponding tree of the truncated R-vine, with Gumbel and t being the dominant chosen copulas.

The detailed results for Trees 3 to 6 of truncated C-vine are provided in the Appendix.

4.5 VaR Performance Evaluation

As mentioned in subsection 3.4, a rolling-window backtesting procedure is used to evaluate the performance of each model. To ensure stability in the estimation of the ARMA–GARCH and the copula modeling, while retaining a sufficiently large out-of-sample data points for meaningful VaR backtesting, a window size of 500 observations is selected.

Table 6: C-vine results for Tree 1

Pair	Family	par	par2	τ
1,30	t	0.20	3.60	0.13
1,29	t	0.84	5.48	0.64
1,28	t	0.67	5.98	0.47
1,27	t	-0.10	3.80	-0.06
1,26	Gumbel	1.23	–	0.19
1,25	t	0.49	8.08	0.33
1,24	Gumbel	1.78	–	0.44
1,23	t	0.09	11.06	0.06
1,22	t	0.21	3.80	0.13
1,21	t	0.36	6.44	0.24
1,20	t	0.41	3.72	0.27
1,19	t	0.78	6.88	0.56
1,18	t	-0.29	3.64	-0.19
1,17	t	-0.25	2.86	-0.16
1,16	t	0.65	5.71	0.45
1,15	t	0.62	4.86	0.43
1,14	t	0.68	5.16	0.48
1,13	t	0.70	5.42	0.50
1,12	t	0.72	7.96	0.52
1,11	t	0.87	7.16	0.67
1,10	t	0.78	4.70	0.57
1,9	t	0.82	5.82	0.61
1,8	t	0.98	12.12	0.89
1,7	t	0.87	4.78	0.68
1,6	Joe	1.41	–	0.19
1,5	t	0.48	5.74	0.32
1,4	t	0.41	4.24	0.27
1,3	t	0.68	6.01	0.47
1,2	t	0.86	13.69	0.65

Table 7: C-vine results for Tree 2

Pair	Family	par	par2	τ
2,30;1	Survival Gumbel	1.14	–	0.12
2,29;1	Frank	1.05	–	0.12
2,28;1	Survival Gumbel	1.03	–	0.03
2,27;1	Survival Joe	1.04	–	0.02
2,26;1	Frank	0.81	–	0.09
2,25;1	t	0.04	8.32	0.02
2,24;1	t	0.26	11.71	0.17
2,23;1	Gumbel	1.03	–	0.03
2,22;1	Survival Gumbel	1.13	–	0.12
2,21;1	Survival Gumbel	1.07	–	0.06
2,20;1	Gaussian	0.17	–	0.11
2,19;1	Frank	1.03	–	0.11
2,18;1	Rotated Joe 90°	-1.06	–	-0.03
2,17;1	t	-0.01	16.19	-0.01
2,16;1	Frank	1.55	–	0.17
2,15;1	Frank	1.41	–	0.15
2,14;1	t	0.54	8.00	0.37
2,13;1	Gaussian	0.44	–	0.29
2,12;1	t	0.75	14.78	0.54
2,11;1	Rotated Clayton 90°	-0.04	–	-0.02
2,10;1	t	0.20	10.49	0.13
2,9;1	t	0.10	11.79	0.07
2,8;1	Rotated Gumbel 90°	-1.12	–	-0.11
2,7;1	t	0.12	9.43	0.08
2,6;1	Survival Clayton	0.08	–	0.04
2,5;1	Survival Joe	1.05	–	0.03
2,4;1	Gaussian	0.24	–	0.16
2,3;1	Survival Gumbel	1.03	–	0.03

This corresponds to approximately 9.6 years of weekly data and covers both crisis and more stable market periods. Given a total of 927 weekly observations, this design yields 427 out-of-sample forecasts.

However, the multivariate t-copula requires approximately four hours to fit a single estimation window, making a full rolling-window backtesting computationally infeasible for this thesis. Therefore, this model is excluded from the rolling-window comparison. As an alternative benchmark, a rolling mean-variance portfolio without short-selling is adopted as a baseline.

Rolling VaR backtesting: MV Portfolio as benchmark As a benchmark specification, portfolio risk is evaluated using a rolling mean-variance portfolio combined with VaR approach. At each rolling estimation date t, the portfolio weights are obtained from a mean–variance optimization problem based on the estimated conditional mean vector $\hat{\boldsymbol{\mu}}_t$ and covariance matrix $\hat{\Sigma}_t$. Specifically, the portfolio is constructed as

$$\hat{\mathbf{w}}_t = \arg\max_{\mathbf{w}} \ \mathbf{w}^\top \hat{\boldsymbol{\mu}}_t \quad \text{s.t. } \mathbf{w}^\top \hat{\Sigma}_t \mathbf{w} \leq \sigma_*^2, \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w} \geq 0.$$

Given these weights, the one-step-ahead conditional portfolio mean and volatility are computed as

$$\hat{\mu}_{p,t+1} = \hat{\mathbf{w}}_t^\top \hat{\boldsymbol{\mu}}_{t+1}, \qquad \hat{\sigma}_{p,t+1} = \sqrt{\hat{\mathbf{w}}_t^\top \hat{\Sigma}_{t+1} \hat{\mathbf{w}}_t}.$$

Assuming conditional normality, the α-level Value-at-Risk forecast is then given by

$$\widehat{\mathrm{VaR}}_{\alpha,t+1} = -\left(\hat{\mu}_{p,t+1} + z_\alpha \hat{\sigma}_{p,t+1}\right).$$

The realized out-of-sample portfolio return is

$$R_{p,t+1}^{\text{real}} = \hat{\mathbf{w}}_t^\top \mathbf{R}_{t+1}^{\text{real}},$$

and a VaR violation occurs whenever

$$R_{p,t+1}^{\text{real}} < -\widehat{\mathrm{VaR}}_{\alpha,t+1}.$$

Repeating this procedure over the out-of-sample period generates a sequence of VaR forecasts and violation indicators, and these subsequently assessed using standard backtesting tools such as the Kupiec unconditional coverage test as mentioned in subsection 3.4

Table 8 shows the backtesting results for the rolling MV portfolio.

Table 8: Kupiec Unconditional Coverage Test for VaR Forecasts–MV

VaR Level	LR_{uc}	p-value	Violations	Sample Size (n)	Observed Rate
95%	7.7653	0.0053	35	427	0.0820
99%	34.1132	5.20×10^{-9}	21	427	0.0492

The Kupiec unconditional coverage results show that the VaR forecasts are rejected at both the 95% and 99% confidence levels. For the 95% VaR, the model produces 35 violations out of 427 observations, corresponding to an observed violation rate of 0.0820, which is largely higher than the expected rate of 0.05. Likewise, for the 99% VaR, there are 21 violations out of 427 observations, giving an observed violation rate of 0.0492, which is also significantly higher than the expected rate of 0.01. The associated LR_{uc} statistics both yield very small p-values, leading to a clear rejection of the null hypothesis of correct unconditional coverage. The results shown in Table 8 have proved the shortcomings of the MV model for underestimating downside risk, since breaches occur more often than the expected level.

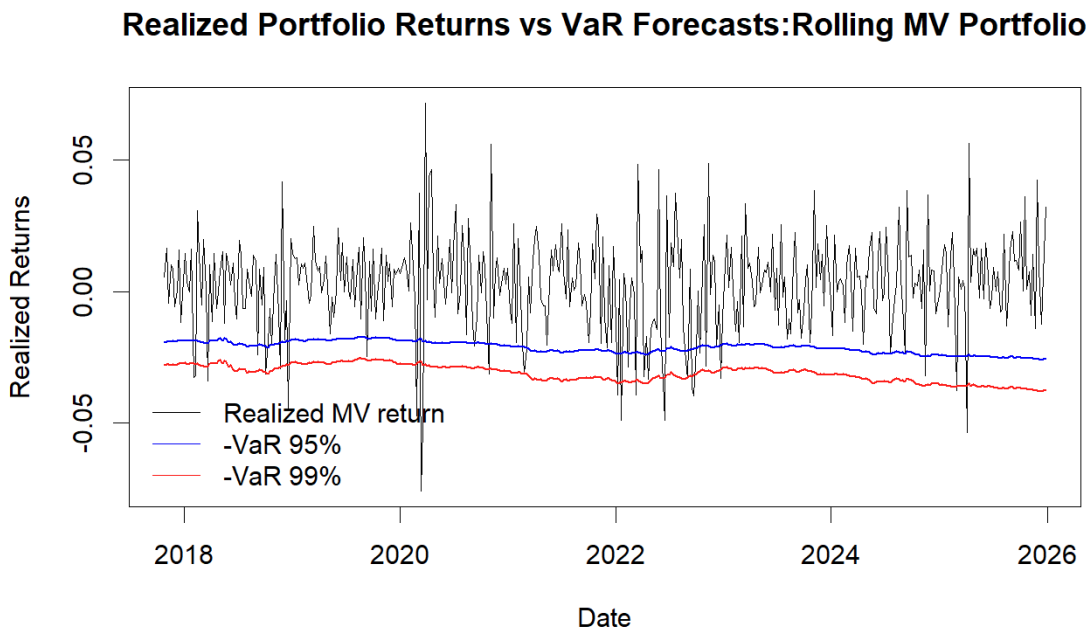

Figure 6: Realized Portfolio Returns and VaR Forecasts:Rolling MV Portfolio

Figure 6 also demonstrates this phenomenon. The realized portfolio returns cross below both the 95% and 99% VaR forecast lines more frequently than expected. The problem is particularly pronounced for the 99% VaR. This shows that the MV portfolio hugely underestimates the tail risks, especially during the period of high volatility where extreme losses would occur.

Rolling VaR backtesting: Truncated R-vine For the truncated R-vine model, it is important to note that the VaR backtesting uses an equally weighted portfolio. The effective sample size is 425 instead of 427, because the GARCH estimation for DBA failed in two rolling windows, potentially due to numerical optimization difficulties.

Table 9: Kupiec Unconditional Coverage Test for VaR Forecasts Truncated R vine

VaR Level	LR_{uc}	p-value	Violations	Sample Size (n)	Observed Rate
95%	3.8908	0.0486	13	425	0.0306
99%	0.0152	0.9020	4	425	0.0094

Table 9 reports the Kupiec test results for the VaR forecasts of the truncated R-vine model. Overall, the results have shown that the model provides a much more satisfactory fit to predict downside risk than in the previous case. At the 95% VaR level, there are 13 violations out of 425 observations, corresponding to an observed exceedance rate of 0.0306. This is below the expected rate of 0.05, indicating that the model slightly overestimates risk, since violations occur less frequently than expected. The associated p-value of 0.0486 is very close to the conventional 5% significance threshold, implying borderline rejection of the null hypothesis of correct unconditional coverage. At the 99% VaR level, the results are much more satisfying. The model records 4 violations out of 425 observations, corresponding to an observed exceedance rate of 0.0094, which is extremely close to the expected rate of 0.01. The Kupiec test statistic is very small, with a p-value of 0.9020, so the null hypothesis of correct unconditional coverage cannot be rejected. This indicates that the truncated R-vine model performs very well in capturing extreme tail risk at the 99% level.

While the 95% VaR may be slightly too conservative, the 99% VaR forecast appears to be very well calibrated. From the risk management perspective, this is an encouraging result, since accurate modeling of the extreme lower tail is particularly important for assessing severe portfolio losses.

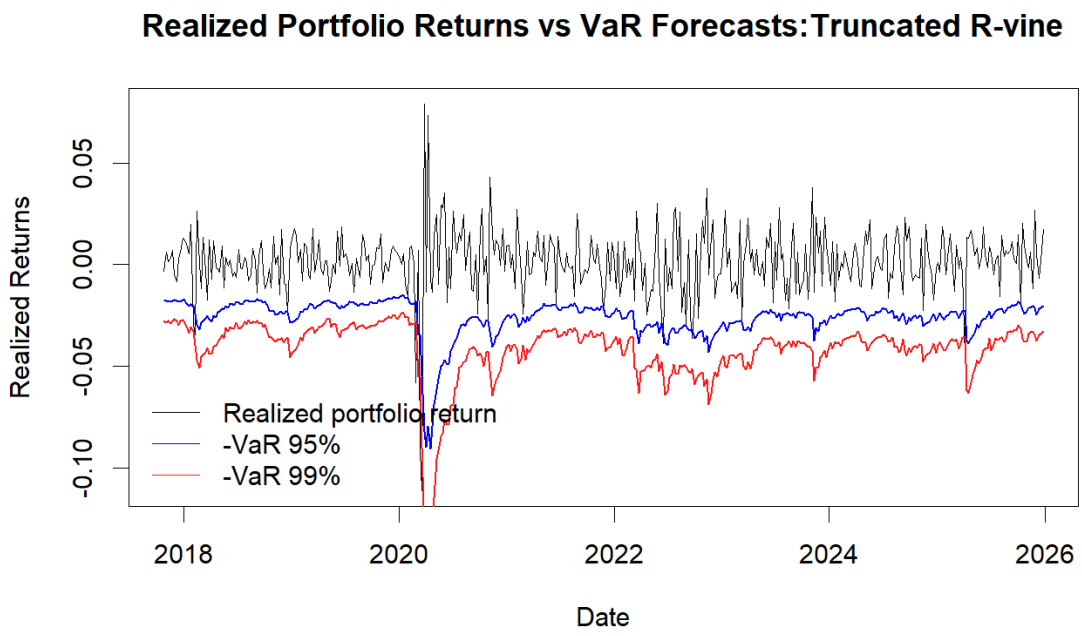

Figure 7: Realized Portfolio Returns and VaR Forecasts:Truncated C-Vine

Figure 7 is a graphical way to show the backtesting results reported in Table 9. The realized portfolio returns only occasionally cross the 95% and 99% VaR forecast lines,. The VaR forecasts also denmonstrate clear time-variant volatility, becoming substantially more negative during periods of market stress, especially around 2020, the Covid period, which suggests that the model is responsive to changes in market conditions and tail risk.

Rolling VaR backtesting: Truncated C-vine The truncated C-vine also uses an equally weighted portfolio for VaR backtesting. The effective sample size is also 425 instead of 427 because of the failure of the GARCH estimation for DBA.

Table 10: Kupiec Unconditional Coverage Test for VaR Forecasts Truncated C vine

VaR Level	LR_{uc}	p-value	Violations	Sample Size (n)	Observed Rate
95%	1.0466	0.3063	26	425	0.0612
99%	0.6454	0.4218	6	425	0.0141

Table 10 reports the Kupiec test results for the VaR forecasts produced by the truncated C-vine model. Overall, the results indicate that the truncated C-vine provides satisfying VaR forecasts at both the 95% and 99% confidence levels with no rejections of the null hypothesis. At the 95% VaR level, there are 26 violations out of 425 observations, corresponding to an observed violation rate of 0.0612. Although this is slightly above the

nominal rate of 0.05, the Kupiec test statistic is 1.0466 with a p-value of 0.3063, so the null hypothesis of correct unconditional coverage cannot be rejected. This suggests that the 95% VaR forecasts are statistically acceptable, even if they slightly underestimate downside risk. At the 99% VaR level, the model records 6 violations out of 425 observations, yielding an observed voilation rate of 0.0141, which is modestly above the nominal level of 0.01. The corresponding Kupiec test statistic is 0.6454 with a p-value of 0.4218, again indicating that the null hypothesis of correct unconditional coverage cannot be rejected.

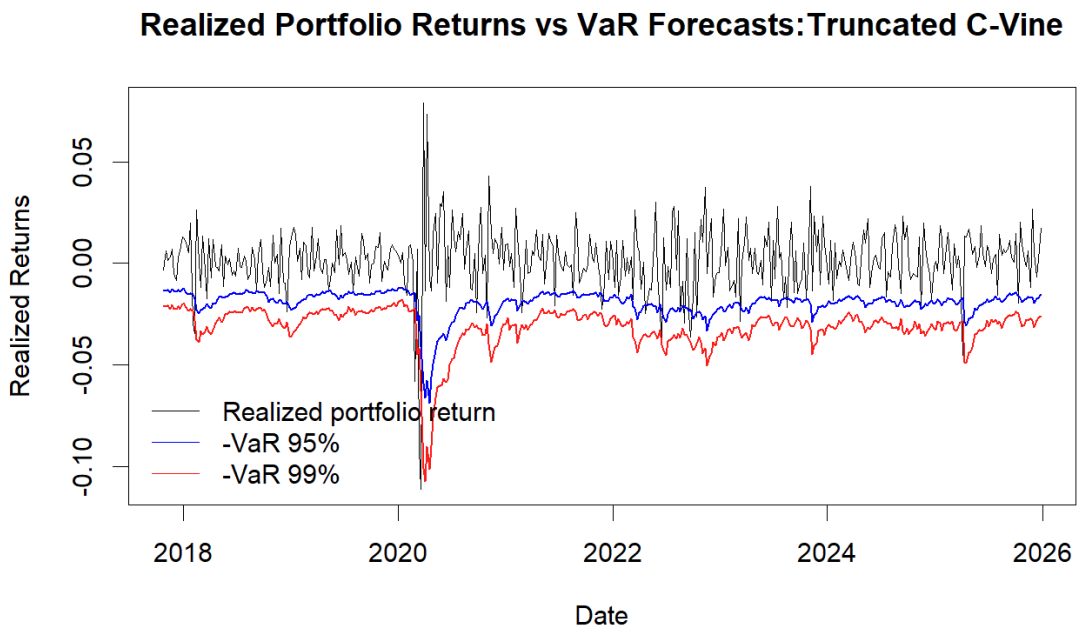

Figure 8: Realized Portfolio Returns and VaR Forecasts:Truncated C-Vine

Figure 8 supports the backtesting results of Table 10 from another angle. The realized portfolio returns generally remain above the 95% and 99% VaR forecast lines, with only occasional violations. The VaR forecasts are clearly time-varying and become substantially more accurate during periods of market crisis, particularly during the Covid period, indicating that the truncated C-vine model is able to adapt to changing market conditions. The sharp downward adjustment of both VaR series during this period is consistent with the heightened volatility and tail risk observed in the realized returns.

All in all, the classical Mean-Variance method clearly leads to the worst performance in terms of VaR prediction. Compared with the truncated R-vine results, the truncated C-vine appears slightly less conservative and more accurate, and thus is the best model in terms of VaR prediction for our analysis.

5 Conclusion

This thesis investigated whether high-dimensional vine copula models can improve portfolio risk predictions.

The empirical results showed that vine copula models were able to capture more complex dependence patterns, and the truncation strategy also helped reduce model complexity while preserving the main dependence structure. In the backtesting analysis, the vine-based approach, particularly truncated C-vine, produced more accurate risk forecasts for the portfolio Value-at-Risk metric.

These findings contribute to the literature by showing that Vine dependence models can be very useful tools for portfolio risk management as compared to the classical Minimum-variance method.

Nevertheless, several limitations should be noted for this analysis. First, the analysis relied on a specific sample period and weekly returns, which may affect the generalizability of the results. Second, the choice of truncation level and pair-copula families may influence model performance. Third, the computational burden of high-dimensional vine, especially the multivariate elliptical copula estimation remains non-negligible.

Future research could extend this framework by considering alternative truncation strategies, higher dimensional asset universes and so on. Another promising direction would be to combine dependence modeling with machine learning techniques for asset selection and portfolio construction.

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Declaration

The author declares that AI tools, including Chatgpt, GitHub Copilot, were used for proofreading and code optimization. These tools were used soley to enhance the clarity of the thesis. All the content of the thesis is original and remains entirely the work of the author, under the guidance of the thesis supervisor and the scientific committee of statistics of University of Geneva.

Appendix

More Theoretical Background

Elliptical Copulas and the Multivariate t-Copula Copulas provide a flexible way to separate marginal behavior from dependence. By Sklar's theorem, for any d-dimensional joint distribution function F with continuous marginals $F_1, \ldots, F_d$, there exists a unique copula C such that

$$F(x_1, \ldots, x_d) = C\big(F_1(x_1), \ldots, F_d(x_d)\big)$$

Hence, the copula fully characterizes the dependence structure among the components of a random vector (Sklar, 1959; Nelsen, 2006; Joe, 2014).

A particularly important class of copulas in finance is the class of *elliptical copulas*, which arise from elliptical distributions such as the multivariate Gaussian and multivariate Student-t distributions. Elliptical copulas are attractive because they provide a tractable multivariate dependence model and naturally extend familiar correlation-based modeling frameworks (McNeil et al., 2015; Joe, 2014). If a random vector $X = (X_1, \ldots, X_d)^\top$ follows an elliptical distribution with continuous marginals, then its copula is called an elliptical copula.

The Gaussian copula is the simplest and most widely known elliptical copula. However, it has an important limitation in risk management applications: it exhibits zero tail dependence whenever the linear correlation is strictly less than one. This means that it cannot capture the tendency of financial assets to experience joint extreme losses or gains (Demarta and McNeil, 2005; McNeil et al., 2015). In contrast, the multivariate t-copula allows for symmetric tail dependence and is therefore often considered a more appropriate benchmark in financial applications.

Let $T_{\nu,R}$ denote the d-dimensional Student-t distribution with correlation matrix R and degrees of freedom ν, and let t_ν denote the univariate Student-t distribution function with ν degrees of freedom. The corresponding t-copula is defined as

$$C^t_{\nu,R}(u_1, \ldots, u_d) = T_{\nu,R}\big(t_\nu^{-1}(u_1), \ldots, t_\nu^{-1}(u_d)\big), \qquad (u_1, \ldots, u_d) \in [0,1]^d.$$

Its density is given by

$$c^t_{\nu,R}(u_1, \ldots, u_d) = \frac{f_{\nu,R}\big(t_\nu^{-1}(u_1), \ldots, t_\nu^{-1}(u_d)\big)}{\prod_{j=1}^d f_\nu\big(t_\nu^{-1}(u_j)\big)},$$

where $f_{\nu,R}$ is the multivariate Student-t density and f_ν is the univariate Student-t density (Demarta and McNeil, 2005; McNeil et al., 2015).

The main advantage of the t-copula over the Gaussian copula is its ability to model tail dependence. For a bivariate t-copula with correlation parameter ρ and degrees of freedom ν, the coefficient of upper tail dependence is

$$\lambda_U = 2\, t_{\nu+1}\left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}}\right),$$

and by symmetry the lower tail dependence coefficient satisfies $\lambda_L = \lambda_U$ (Demarta and McNeil, 2005). Therefore, the t-copula is capable of capturing joint extreme movements in both tails, which makes it especially useful in portfolio risk modeling.

Nevertheless, elliptical copulas remain globally symmetric and may still be too restrictive in high-dimensional financial applications. In particular, they impose a single dependence class on all pairs of variables and cannot correct model heterogeneous pairwise structures, asymmetric dependence, or varying tail behavior across different asset pairs. This motivates the use of more flexible constructions based on bivariate building blocks, namely pair-copula constructions.

Pair-Copula Constructions Vine copula was first introduced with the theory of Pair-copula constructions (PCCs) proposed by Aas et al. (2009), and it provides the fundamental framework for decomposing a multivariate copula into products of bivariate copulas. In general, the key idea of this framework is to factorize a multivariate density into marginal densities and a product of conditional bivariate copula densities. This allows each pairwise dependence relationship to be modeled separately, each pairwise dependecen thus can be modeled with a different copula family.

To illustrate the idea, Aas et al. (2009) give an example in three dimensions. The joint density of a three-dimensional random vector (X_1, X_2, X_3) with continuous marginals can be written as

$$f(x_1, x_2, x_3) = f_1(x_1) f_2(x_2) f_3(x_3)\, c_{12}(u_1, u_2)\, c_{23}(u_2, u_3)\, c_{13|2}(u_{1|2}, u_{3|2}),$$

where $u_j = F_j(x_j)$, c_{12} and c_{23} are unconditional pair-copula densities, and $c_{13|2}$ is the copula density of (X_1, X_3) conditional on X_2 (Aas et al., 2009). The terms $u_{1|2}$ and $u_{3|2}$ denote conditional distribution values, typically obtained through the h-functions. where the h-function is defined as

$$h_{1|2}(u_1 \mid u_2) = \frac{\partial C_{12}(u_1, u_2)}{\partial u_2}$$

More generally, for a d-dimensional random vector, a pair-copula construction expresses the copula density as a product of bivariate copula densities organized through a vine

structure.

One of the biggest advantage of Pair Copula Constructions is the flexibility it allows. Unlike multivariate elliptical copulas or multivariate Archimedean copulas, they do not impose one global dependence structure on all variables. Instead, different copula families can be chosen for different pairs, allowing the model to capture many kinds of different dependence structures, including asymmetric dependence, different tail features, and heterogeneous strengths of association. This flexibility is particularly valuable in high-dimensional financial applications, where dependence patterns are often complex and differ substantially across asset classes (Aas et al., 2009; Joe, 2014).

Independence Copula An important benchmark in copula theory is the *independence copula*. If $U_1, \ldots, U_d$ are mutually independent random variables uniformly distributed on $[0, 1]$, then their copula is

$$C^{\perp}(u_1, \ldots, u_d) = \prod_{j=1}^{d} u_j, \qquad (u_1, \ldots, u_d) \in [0, 1]^d$$

Its density is simply

$$c^{\perp}(u_1, \ldots, u_d) = 1$$

Thus, the independence copula represents the absence of dependence among the components (Nelsen, 2006; Joe, 2014).

Although seemingly trivial, the independence copula plays an important theoretical and practical role. First, it serves as a reference model against which more complex dependence structures can be compared. Second, in vine copula modeling it is commonly used as a truncation device: if dependence in higher-order trees is weak, the corresponding pair-copulas may be replaced by the independence copula, thereby reducing model complexity and the number of parameters (Aas et al., 2009; Joe, 2014). This idea is particularly useful in high-dimensional settings, where a full vine specification may be unnecessarily rich.

Moreover, copula family selection procedures often include the independence copula as one of the candidate models. If the data do not provide evidence for a significant dependence relationship for a particular pair, selecting the independence copula avoids overfitting and improves parsimony.

Copula Simulation Simulation from a fitted copula model is crucial to check the model performance because it allows one to generate artificial dependence-consistent scenarios and to approximate quantities such as portfolio loss distributions and Value-at-Risk

(McNeil et al., 2015; Joe, 2014). The general principle is to simulate a random vector

$$(U_1, \ldots, U_d) \sim C,$$

where C is the fitted copula, and then transform the simulated uniforms back to the original marginal scale using the inverse marginal distribution functions.

For elliptical copulas, simulation is relatively straightforward. In the case of the Gaussian copula, one first simulates a multivariate normal random vector $Z \sim N_d(0, R)$, then applies $U_j = \Phi(Z_j), \qquad j = 1, \ldots, d$. The resulting vector $(U_1, \ldots, U_d)$ follows the Gaussian copula with correlation matrix R (McNeil et al., 2015).

For the multivariate t-copula, simulation proceeds similarly but includes an additional scale mixture step. One may simulate

$Y \sim N_d(0, R), \qquad S \sim \chi^2_\nu$, independently, and define

$X = \sqrt{\frac{\nu}{S}}\, Y$ .Then X follows a multivariate Student-t distribution with correlation matrix R and degrees of freedom ν, and the transformed variables

$$U_j = t_\nu(X_j), \qquad j = 1, \ldots, d,$$

follow the multivariate t-copula (Demarta and McNeil, 2005; McNeil et al., 2015) where the tail dependence is preserved by the fitted t-copula.

For pair-copula constructions, as mentioned above, the dependence structure is built recursively from conditional bivariate copulas. Thus, simulation is typically performed sequentially using inverse conditional distribution functions, expressed through inverse h-functions (Aas et al., 2009; Joe, 2014). The inverse h-function, denoted by $h^{-1}_{i|j}$, is used to recover u_i from a conditional probability level. More generally, for a conditional pair-copula $C_{ij|D}$, one writes

$$h_{i|j;D}(u_i \mid u_j, u_D) = \frac{\partial C_{ij|D}(u_i, u_j \mid u_D)}{\partial u_j}.$$

The general simulation procedure can be ffound in the methodology part. Simulation-based evaluation is straightforward once the vine structure, pair-copula families, and parameters have been estimated.

For example, in a three-dimensional C-vine with copulas C_{12}, C_{23}, and $C_{13|2}$, one may first set

$$U_2 = W_2,$$

then generate

$$U_1 = C_{1|2}^{-1}(W_1 \mid U_2), \qquad U_3 = C_{3|2}^{-1}(W_3 \mid U_2),$$

with the final dependence adjustment obtained through the conditional copula $C_{13|2}$. In higher dimensions, this idea generalizes through repeated use of h-functions and inverse h-functions (Aas et al., 2009). Although more computationally demanding than simulation from elliptical copulas, vine simulation offers much greater flexibility because each pair-copula can be chosen separately.

After simulating copula data on the uniform scale, one maps the uniforms back to the marginal distribution through the inverse of the CDF $X_j = F_j^{-1}(U_j), \qquad j = 1, \ldots, d,$ where F_j^{-1} denotes the inverse marginal distribution function.

Overall, copula simulation is a key bridge between statistical dependence modeling and practical applications such as Monte Carlo estimation of portfolio risk measures. In the present thesis, it provides the mechanism through which fitted dependence models can be translated into simulated joint return scenarios and, ultimately, into forecasts of portfolio tail risk.

Truncated R-vine and C-vine outputs

Table 11: R-vine Results for Tree 3

Pair	Family	par	par2	τ
(21,18;8,6)	Survival Gumbel	1.15	–	0.13
(6,19;8,21)	Joe	1.06	–	0.03
(7,21;6,8)	Frank	-0.50	–	-0.06
(9,8;7,6)	Rotated Clayton 270°	-0.10	–	-0.05
(6,5;4,10)	Gaussian	-0.25	–	-0.16
(10,7;9,6)	t	-0.15	9.38	-0.09
(9,4;6,10)	Clayton	0.07	–	0.03
(25,9;10,6)	t	-0.09	11.00	-0.06
(26,24;25,22)	Gaussian	0.13	–	0.08
(20,26;22,25)	t	0.12	9.43	0.08
(17,22;20,25)	Gaussian	0.16	–	0.10
(25,28;20,17)	Joe 90°	-1.03	–	-0.02
(6,17;25,20)	Gumbel 90°	-1.09	–	-0.08
(10,20;6,25)	Gaussian	-0.02	–	-0.01
(10,3;1,29)	Clayton	0.12	–	0.05
(1,25;10,6)	Gumbel 270°	-1.11	–	-0.10
(14,27;29,30)	Gumbel 90°	-1.06	–	-0.06
(6,2;10,1)	Survival Gumbel	1.12	–	0.11
(29,6;1,10)	Frank	0.70	–	0.08
(14,10;29,1)	Frank	1.65	–	0.18
(13,23;12,11)	Clayton 270°	-0.05	–	-0.02
(16,12;13,11)	t	-0.13	11.63	-0.08
(15,11;16,13)	t	-0.07	9.59	-0.04
(14,13;15,16)	t	0.06	5.25	0.04
(29,16;14,15)	Gumbel 270°	-1.03	–	-0.03
(30,15;29,14)	t	0.25	12.57	0.16
(30,1;14,29)	Gaussian	0.14	–	0.09

Table 12: R-vine Results for Tree 4

Pair	Family	par	par2	τ
(19,18;21,8,6)	Frank	0.92	–	0.10
(7,19;6,8,21)	Clayton 90°	-0.03	–	-0.02
(9,21;7,6,8)	t	0.02	13.51	0.01
(10,8;9,7,6)	Gaussian	-0.01	–	-0.01
(9,5;6,4,10)	Frank	0.34	–	0.04
(4,7;10,9,6)	Gumbel 270°	-1.06	–	-0.05
(25,4;9,6,10)	Joe 270°	-1.05	–	-0.03
(1,9;25,10,6)	Frank	0.28	–	0.03
(20,24;26,25,22)	Survival Joe	1.03	–	0.02
(17,26;20,22,25)	Clayton	0.04	–	0.02
(28,22;17,20,25)	t	-0.01	18.80	-0.01
(6,28;25,20,17)	Gumbel 90°	-1.01	–	-0.01
(10,17;6,25,20)	Gumbel 270°	-1.08	–	-0.07
(1,20;10,6,25)	Gumbel 90°	-1.07	–	-0.06
(6,3;10,1,29)	Clayton 270°	-0.16	–	-0.07
(2,25;1,10,6)	Clayton 90°	-0.09	–	-0.04
(1,27;14,29,30)	Gaussian	-0.18	–	-0.12
(29,2;6,10,1)	Clayton	0.10	–	0.05
(14,6;29,1,10)	Survival Gumbel	1.06	–	0.05
(30,10;14,29,1)	Joe	1.05	–	0.03
(16,23;13,12,11)	Gumbel	1.04	–	0.04
(15,12;16,13,11)	t	-0.17	7.45	-0.11
(14,11;15,16,13)	t	-0.48	21.83	-0.32
(29,13;14,15,16)	Gaussian	-0.27	–	-0.17
(30,16;29,14,15)	Frank	0.44	–	0.05
(1,15;30,29,14)	Gumbel	1.03	–	0.03

Table 13: R-vine Results for Tree 5

Pair	Family	par	par2	τ
(7,18;19,21,8,6)	Clayton	0.03	–	0.01
(9,19;7,6,8,21)	Frank	0.15	–	0.02
(10,21;9,7,6,8)	Frank	0.58	–	0.06
(4,8;10,9,7,6)	Frank	0.31	–	0.03
(25,5;9,6,4,10)	Gaussian	-0.14	–	-0.09
(25,7;4,10,9,6)	Gaussian	-0.03	–	-0.02
(1,4;25,9,6,10)	Rotated Joe 270°	-1.08	–	-0.04
(2,9;1,25,10,6)	Rotated Gumbel 270°	-1.02	–	-0.02
(17,24;20,26,25,22)	Rotated Joe 90°	-1.01	–	0.00
(28,26;17,20,22,25)	Frank	0.10	–	0.01
(6,22;28,17,20,25)	t	0.00	10.15	0.00
(10,28;6,25,20,17)	Rotated Clayton 270°	-0.03	–	-0.02
(1,17;10,6,25,20)	Gaussian	-0.11	–	-0.07
(2,20;1,10,6,25)	Rotated Clayton 270°	-0.03	–	-0.02
(2,3;6,10,1,29)	t	-0.17	20.12	-0.11
(29,25;2,1,10,6)	Joe	1.02	–	0.01
(10,27;1,14,29,30)	Clayton	0.04	–	0.02
(14,2;29,6,10,1)	Gaussian	-0.05	–	-0.03
(30,6;14,29,1,10)	Rotated Clayton 90°	-0.05	–	-0.03
(15,10;30,14,29,1)	t	0.21	10.53	0.14
(15,23;16,13,12,11)	Rotated Joe 90°	-1.03	–	-0.02
(14,12;15,16,13,11)	Gaussian	-0.23	–	-0.15
(29,11;14,15,16,13)	Rotated Gumbel 270°	-1.14	–	-0.12
(30,13;29,14,15,16)	Frank	1.37	–	0.15
(1,16;30,29,14,15)	Survival Clayton	0.03	–	0.02

Table 14: R-vine Results for Tree 6

Pair	Family	par	par2	τ
(9,18;7,19,21,8,6)	Rotated Clayton 270°	-0.10	–	-0.05
(10,19;9,7,6,8,21)	Frank	-0.11	–	-0.01
(4,21;10,9,7,6,8)	t	0.09	13.47	0.06
(25,8;4,10,9,7,6)	Frank	0.61	–	0.07
(1,5;25,9,6,4,10)	Rotated Clayton 90°	-0.04	–	-0.02
(1,7;25,4,10,9,6)	Rotated Gumbel 90°	-1.04	–	-0.04
(2,4;1,25,9,6,10)	Rotated Clayton 270°	-0.08	–	-0.04
(29,9;2,1,25,10,6)	Survival Joe	1.03	–	0.02
(28,24;17,20,26,25,22)	Frank	0.26	–	0.03
(6,26;28,17,20,22,25)	Clayton	0.04	–	0.02
(10,22;6,28,17,20,25)	Gaussian	0.23	–	0.15
(1,28;10,6,25,20,17)	Survival Clayton	0.01	–	0.01
(2,17;1,10,6,25,20)	Rotated Joe 270°	-1.02	–	-0.01
(29,20;2,1,10,6,25)	Frank	0.41	–	0.05
(14,3;2,6,10,1,29)	t	-0.04	8.16	-0.02
(14,25;29,2,1,10,6)	Gaussian	0.09	–	0.05
(6,27;10,1,14,29,30)	Clayton	0.03	–	0.01
(30,2;14,29,6,10,1)	Frank	-0.97	–	-0.11
(15,6;30,14,29,1,10)	Gaussian	0.21	–	0.13
(16,10;15,30,14,29,1)	Rotated Clayton 270°	-0.03	–	-0.01
(14,23;15,16,13,12,11)	Rotated Clayton 90°	-0.03	–	-0.01
(29,12;14,15,16,13,11)	Gaussian	-0.05	–	-0.03
(30,11;29,14,15,16,13)	Frank	0.70	–	0.08
(1,13;30,29,14,15,16)	Rotated Gumbel 270°	-1.09	–	-0.09

Table 15: C-vine Results for Tree 3

Pair	Family	par	par2	τ
3,30;2,1	t	0.07	8.65	0.05
3,29;2,1	t	0.43	7.12	0.29
3,28;2,1	t	0.99	4.82	0.92
3,27;2,1	Frank	0.44	–	0.05
3,26;2,1	Rotated Clayton 90°	-0.13	–	-0.06
3,25;2,1	t	-0.05	9.87	-0.03
3,24;2,1	Survival Joe	1.06	–	0.03
3,23;2,1	Frank	0.23	–	0.03
3,22;2,1	t	0.08	9.57	0.05
3,21;2,1	Rotated Joe 270°	-1.03	–	-0.02
3,20;2,1	Frank	-0.48	–	-0.05
3,19;2,1	t	0.35	9.30	0.23
3,18;2,1	Survival Gumbel	1.19	–	0.16
3,17;2,1	t	0.29	11.35	0.19
3,16;2,1	t	0.33	9.08	0.21
3,15;2,1	t	0.33	9.19	0.21
3,14;2,1	Clayton	0.09	–	0.04
3,13;2,1	Clayton	0.15	–	0.07
3,12;2,1	Rotated Clayton 270°	-0.10	–	-0.05
3,11;2,1	t	-0.03	9.99	-0.02
3,10;2,1	Gaussian	0.24	–	0.15
3,9;2,1	t	0.41	7.46	0.27
3,8;2,1	t	0.01	7.58	0.01
3,7;2,1	Gaussian	0.39	–	0.26
3,6;2,1	t	0.38	10.13	0.25
3,5;2,1	Rotated Gumbel 270°	-1.06	–	-0.06
3,4;2,1	t	0.04	5.76	0.03

Table 16: C-vine Results for Tree 4

Pair	Family	par	par2	τ
4,30;3,2,1	t	0.81	16.55	0.60
4,29;3,2,1	Frank	-0.56	–	-0.06
4,28;3,2,1	t	0.02	13.98	0.01
4,27;3,2,1	Frank	1.72	–	0.19
4,26;3,2,1	t	0.20	7.53	0.13
4,25;3,2,1	t	0.22	14.32	0.14
4,24;3,2,1	Survival Gumbel	1.21	–	0.18
4,23;3,2,1	t	-0.02	15.00	-0.01
4,22;3,2,1	t	0.82	16.15	0.61
4,21;3,2,1	Clayton	0.22	–	0.10
4,20;3,2,1	Clayton	0.27	–	0.12
4,19;3,2,1	t	-0.02	9.12	-0.01
4,18;3,2,1	Frank	0.74	–	0.08
4,17;3,2,1	Frank	1.13	–	0.12
4,16;3,2,1	t	0.16	10.86	0.10
4,15;3,2,1	Gumbel	1.09	–	0.08
4,14;3,2,1	Frank	0.71	–	0.08
4,13;3,2,1	Rotated Gumbel 90°	-1.03	–	-0.03
4,12;3,2,1	Frank	-0.44	–	-0.05
4,11;3,2,1	Rotated Gumbel 90°	-1.09	–	-0.08
4,10;3,2,1	Gaussian	-0.16	–	-0.10
4,9;3,2,1	Gaussian	-0.09	–	-0.06
4,8;3,2,1	Survival Gumbel	1.10	–	0.09
4,7;3,2,1	Gaussian	-0.19	–	-0.12
4,6;3,2,1	Frank	0.63	–	0.07
4,5;3,2,1	Gaussian	0.29	–	0.19

Table 17: C-vine Results for Tree 5

Pair	Family	par	par2	τ
5,30;4,3,2,1	Gaussian	0.20	–	0.13
5,29;4,3,2,1	t	-0.15	17.47	-0.10
5,28;4,3,2,1	Survival Joe	1.02	–	0.01
5,27;4,3,2,1	t	0.41	9.60	0.27
5,26;4,3,2,1	t	0.30	9.49	0.19
5,25;4,3,2,1	t	0.60	15.25	0.41
5,24;4,3,2,1	t	0.45	9.30	0.30
5,23;4,3,2,1	Clayton	0.07	–	0.04
5,22;4,3,2,1	Gaussian	0.22	–	0.14
5,21;4,3,2,1	t	0.05	10.36	0.03
5,20;4,3,2,1	t	-0.04	11.72	-0.02
5,19;4,3,2,1	Frank	0.10	–	0.01
5,18;4,3,2,1	t	0.14	12.77	0.09
5,17;4,3,2,1	Gumbel	1.14	–	0.12
5,16;4,3,2,1	Frank	0.85	–	0.09
5,15;4,3,2,1	Frank	0.66	–	0.07
5,14;4,3,2,1	Frank	0.25	–	0.03
5,13;4,3,2,1	Rotated Gumbel 90°	-1.05	–	-0.04
5,12;4,3,2,1	Rotated Clayton 270°	-0.07	–	-0.03
5,11;4,3,2,1	Gaussian	-0.10	–	-0.06
5,10;4,3,2,1	Rotated Gumbel 270°	-1.09	–	-0.08
5,9;4,3,2,1	Rotated Gumbel 270°	-1.09	–	-0.08
5,8;4,3,2,1	Gaussian	0.13	–	0.09
5,7;4,3,2,1	t	-0.21	15.49	-0.14
5,6;4,3,2,1	Survival Gumbel	1.09	–	0.08

Table 18: C-vine Results for Tree 6

Pair	Family	par	par2	τ
6,30;5,4,3,2,1	t	0.15	10.11	0.10
6,29;5,4,3,2,1	Gaussian	-0.12	–	-0.08
6,28;5,4,3,2,1	t	-0.03	12.49	-0.02
6,27;5,4,3,2,1	Frank	2.13	–	0.23
6,26;5,4,3,2,1	Clayton	0.09	–	0.04
6,25;5,4,3,2,1	Survival Gumbel	1.05	–	0.04
6,24;5,4,3,2,1	Survival Joe	1.06	–	0.03
6,23;5,4,3,2,1	Rotated Joe 270°	-1.09	–	-0.05
6,22;5,4,3,2,1	Gumbel	1.11	–	0.10
6,21;5,4,3,2,1	Rotated Clayton 90°	-0.12	–	-0.05
6,20;5,4,3,2,1	Rotated Gumbel 270°	-1.10	–	-0.09
6,19;5,4,3,2,1	t	0.46	17.45	0.30
6,18;5,4,3,2,1	t	0.77	6.69	0.56
6,17;5,4,3,2,1	t	0.78	7.61	0.57
6,16;5,4,3,2,1	t	0.62	6.10	0.42
6,15;5,4,3,2,1	t	0.62	7.12	0.43
6,14;5,4,3,2,1	Clayton	0.03	–	0.02
6,13;5,4,3,2,1	Rotated Gumbel 90°	-1.00	–	0.00
6,12;5,4,3,2,1	Rotated Gumbel 270°	-1.04	–	-0.04
6,11;5,4,3,2,1	Survival Gumbel	1.07	–	0.07
6,10;5,4,3,2,1	Rotated Clayton 270°	-0.01	–	-0.01
6,9;5,4,3,2,1	Frank	-0.94	–	-0.10
6,8;5,4,3,2,1	Frank	-0.56	–	-0.06
6,7;5,4,3,2,1	Frank	-0.60	–	-0.07